Expensive coffee, richer farmers?

Understanding prices, quantities and producer revenue

Economic Analysis for Data Scientists – Pau Milán (BSE)

Your assignment

You are a data scientist advising an organisation representing coffee producers. Coffee prices fluctuate, and the organisation wants to understand why. Do higher prices mean stronger demand? Do they mean that producers are better off? Your task is to investigate the evidence, explain market outcomes using Unit 4, and use your analysis to predict the consequences of a smaller available harvest and a tax on coffee sales.

Work in **groups of three**. The questions organise the investigation; they do not prescribe an analysis recipe. You decide which calculations, tables and figures provide the strongest evidence, and explain why each belongs in your report.

Research context. This exercise is inspired by Andrea Bastianin, Alessandro Lanza and Matteo Manera (2018), "Economic impacts of El Niño Southern Oscillation: evidence from the Colombian coffee market", *Agricultural Economics*, 49(5), 623–633. The paper investigates weather, supply and demand as sources of coffee-market fluctuations, and why the consequences of a price change depend on its cause. Our data are **synthetic teaching data**, not observations from their study; the market assumptions and numerical results differ. We do not replicate their econometric methods. Reading the article is not required. Published article; open manuscript.

The setting and data

In `coffee_market.csv`, one row describes a selling period in a stylised wholesale market for one standardised grade of green coffee, sold to foreign buyers. There are **300 independent observations**. These are fictional selling periods, not actual calendar months or individual farms. Prices and quantities refer to the same product and market.

Variable(s)	Description
`period_id`, `period_index`	Observation identifier and display order; the index is not a historical trend.
`price_usd_kg`	Wholesale price, US dollars per kilogram.
`quantity_kt`	Quantity traded, thousand tonnes per selling period.
`growing_conditions`	Conditions in the producing region during the preceding growing window: -1.5 unfavourable, 0 normal, 1.5 favourable.
`buyer_activity`	Economic activity among foreign buyers; zero denotes reference conditions.

Assume price-taking buyers and sellers, market clearing and fixed quality. Growing conditions are independent of foreign buyers' preferences, incomes and outside options, including unobserved demand disturbances. Buyer activity does not directly change producers' costs. Over these selling periods, sellers can adjust harvesting, processing and sales effort; there is no inventory carry-over. Explain where your analysis uses these assumptions.

One thousand tonnes equals one million kilograms: price times quantity in the supplied units gives sales revenue in **million US dollars**. Additional files describe two market episodes and new forecast scenarios; use them when directed below. Their columns and units are documented in `data/README.md`. **Do not open** `coffee_forecast_reveal.csv` **before recording your predictions in Question 5.**

Explore, explain and measure

1 Understand and explore the market

Explore and audit `coffee_market.csv`. Give a concise account of the sample, units and any data issues. Investigate prices, quantities and aggregate sales revenue, choosing a small set of descriptive tables and figures that makes the important patterns clear.

What relationship do you observe between prices and quantities? What does it tell you about the market, and what does it leave unresolved? In particular, does it establish how buyers would respond if the price of coffee increased? Explain why you selected each piece of evidence and identify the questions your exploration raises.

2 Develop an equilibrium explanation

Use the supply-and-demand framework from Unit 4 to make sense of your findings. Explain what an observed price–quantity pair represents in this framework. Develop alternative explanations for a higher coffee price, and use clearly labelled diagrams to show how their predictions differ. Distinguish movements along a curve from shifts of a curve.

What comparison would allow you to learn how buyers respond to prices? Propose a comparison using the available information and explain why it is informative. State what changes across the observations you compare, what needs to remain comparable, and which assumptions justify your interpretation.

Demand may fluctuate for reasons you do not observe. Explain whether, and under what conditions, your proposed comparison can still be useful. You do not need to estimate preferences or a production function; the essential task is to connect the source of variation in the data to the equilibrium model.

3 Measure the demand response

Implement the comparison you proposed and present evidence that makes its logic visible. Estimate the size of the demand response both in quantity units and as a price elasticity. State the point or range at which you evaluate the elasticity and explain its economic meaning. Compare your answer with the relationship obtained by treating all price–quantity observations alike.

Investigate whether differences in observed buyer conditions affect your interpretation. What evidence would strengthen your argument, and what would cast doubt on it? Distinguish what you can check in the data from what you must assume.

A linear approximation to demand over the observed range is acceptable. Explain how you obtain it and why it answers the economic question. Discuss the limitations and uncertainty of your estimate in plain language. Formal instrumental-variables estimation and confidence intervals are not required; you may use the supplied resampling helper if useful.

Preparing for the TA session
Bring an initial exploration and an attempted answer to Question 2, including your proposed comparison. The session will help you connect your evidence to the theory and develop the remaining investigation; you do not need to have solved the whole assignment beforehand.

Interpret, predict and advise

4 Investigate two price increases

Open `coffee_episodes.csv`. It contains two separate episodes, A and B, each with ten observations before and ten during a change. Compare prices, quantities, revenue and market conditions. Do not add these observations to your demand-estimation sample.

Explain each episode using Unit 4 and evidence from the data. Can similar price increases have different causes and consequences for sellers? Use your estimated demand elasticity where appropriate, explaining when it applies. Does your evidence establish that profits increased, that every farmer benefited, or that society became better off? Identify what further information stronger claims would require.

5 Predict a new market outcome

Open `coffee_forecast_scenarios.csv`. These scenarios concern a **shorter selling window after harvest**: the coffee is already harvested, graded and committed to sale. Normal available volume is 100 thousand tonnes; the other scenario has 8% less. All available coffee must be sold, with no storage, replacement imports, quality changes or alternative outlets. Buyer activity is at its reference level, other demand disturbances average zero, and your recovered demand relationship remains valid.

Using only your historical-sample estimates, predict the market-clearing price and total sales revenue in each scenario. Show how the supply constraint and demand jointly determine your prediction, using a diagram and calculations. What would the original pooled price–quantity relationship predict instead? If the available harvest falls but the price initially remains unchanged, how do desired purchases compare with available supply? Explain how the new equilibrium resolves this discrepancy.

Record your predictions before opening `coffee_forecast_reveal.csv`. This file contains 40 independent realisations of each scenario. Compare your predictions with the new evidence without refitting your model. Explain any discrepancies and distinguish predicting an average from predicting one observation.

6 Consider a tax on sales

Return to the **normal-volume scenario** in Question 5. The government introduces a tax of $1 per kilogram, paid by sellers on every kilogram sold. All other assumptions of that scenario remain in place.

Use the tax framework from Unit 4 to predict the price buyers pay, the price sellers receive after tax, and the quantity traded. Calculate government revenue and the change in aggregate seller receipts after tax. Illustrate the equilibrium and explain who bears the tax. Does it create a deadweight loss in this setting? Use the standard surplus framework from class, assuming no externalities or tax-collection costs. Would your answer change if buyers, rather than sellers, were legally responsible for paying the same tax? Explain using the prices relevant to each side of the market. Which assumption drives your incidence result, and why might it differ over a longer horizon?

7 Brief the producers' organisation

Write a **200–300 word briefing**: what explains price movements, when can higher prices raise seller revenue, and what do your harvest and tax scenarios imply? Explain the central reasoning in plain language, distinguish evidence from assumptions, and identify the most useful additional information before offering broader advice.

What to hand in

Submit a self-contained **R Jupyter notebook** combining code, selected outputs and written reasoning. Include your economic diagrams (digital or clear photographs), recorded predictions and final briefing. It should run from the supplied CSVs. Label figures and tables, give units and sample sizes, and explain how each output supports a claim.

Your work will be assessed on the justification of your comparisons, the coherence of your economic reasoning, the quality of your evidence, and clarity and reproducibility. There is no required set of plots or unique presentation of the answer. **You are not required to estimate supply.**